Benjamin Shvartsman

717-379-9809 — b.shvartsman@columbia.edu — linkedin.com/in/bshvartsman — github.com/frogsterr

Education

Columbia University — New York City, NY
B.A in Computer Science, Jewish History, Biology — *Expected Grad: May 2028*

- **GPA: 3.9 / 4.0**; Dean's List
- **Awards:** ICPC, Amazon Future Engineer Scholar (1 of 400 National), List College Fellow
- **Coursework:** Data Structures & Algorithms, ML for Functional Genomics, Advanced Systems Programming, Computer Architecture, Linear Algebra, Probability, Calculus III

Experience

YouTube — Mountain View, CA
Machine Learning Software Engineer Intern — *May 2026 – Current*

- Co-architected a Transformer and XGBoost hybrid classification pipeline for copyright spam detection with **99% precision**, orchestrating inference for **1M+ weekly** claims within a high-throughput C++ takedown aggregator.
- Refactored the end-to-end TensorFlow feature registration workflow within the C++ scheduler, **reducing code footprint by 40–60%** and heavily optimizing downstream ingestion pipeline latencies.
- Engineered an Automatic Prompt Optimization (APO) pipeline using an **NSGA-II evolutionary algorithm** and Pareto ranking for dynamic prompt topology selection, improving safety and accuracy by **18%**.

Columbia Creative Machines Labratory — New York, NY
Machine Learning Research Intern — *May 2026 – Current*

- Formulated a mechanistic interpretability framework to track topological drift of safety guardrails during continual multilingual training of Transformers
- Optimized distributed GPU cluster training utilizing **GaLore** and mixed-precision (BF16), writing **custom CUDA and Triton kernels** to maximize hardware utilization and accelerate the training loop for 1B+ parameter models.

NVIDIA — New York, NY
ML Systems Engineer Intern, Research — *January 2026 – April 2026*

- Pioneered neuro-symbolic causal graph reasoning for GPU kernel crash diagnosis by modeling cross-subsystem dependencies and defect patterns; **35% faster root cause resolution** over baseline embedding-based retrieval.
- Discovered upwards of **50× token efficiency gains** for tool-augmented Transformer code reasoning by architecting a gRPC-based LLVM/Clang semantic index server — replaced $O(n)$ unstructured retrieval with $O(1)$ graph-structured LSP queries (**20M+ tokens/month** saved).
- Achieved **9× startup speedup** for internal ML tooling via lazy import refactoring and OS-level caching, diagnosed through multi-layer profiling (cProfile, Nsight, Windows Performance Analyzer).

Google — Sunnyvale, CA
Software Engineer Intern — *May 2025 – August 2025*

- Led cross-team development of an agentic investigation system for SREs on GCP's Agent Development Kit—projected **$150K annual savings** and 100+ developer hours saved.
- Fine-tuned Gemini via LoRA and knowledge distillation, achieved **22% improvement** in reasoning accuracy.
- Built distributed TPU training pipeline, processing 2,000+ bug tickets per weekly cycle.
- Optimized similarity search for RAG via vector approximation algorithms to achieve **4× faster query response**.

Technical Skills & Interests

Programming Languages: C/C++, Python, CUDA, Triton, Java, Go, SQL, R, TypeScript
Frameworks & Libraries: PyTorch, JAX, Pandas, DeepSpeed, NumPy, Spark, FastAPI, OpenCV
Developer Tools: Git, Docker, Linux (Nsight, cProfile), AWS, GCP, Azure, HPC Clusters
Interests: Ironman Triathlon, Squash, Quantitative Finance, Classical Piano, Tennis